

WEAVE: Affordable Real Style Transfer via Wavelet-Endpoint Aligned Vector-Field Editing

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Abstract

Current style transfer faces two main issues. First, state-of-the-art diffusion models and their training-free variants demand massive GPU memory, rendering them impractical for consumer hardware. Second, lightweight alternatives often underperform a trivial identity mapping, which scores high on traditional metrics but was rarely evaluated as a baseline. Therefore, an effective method should outperform the identity baseline while remaining computationally affordable. We hypothesize that this failure stems from frequency competition in shared latent spaces, where optimizing for strong structural signals suppresses subtle stylistic details.

To address this, we propose WEAVE, which uses wavelet de-

composition to separate frequencies. We operate in the VAE latent space for efficiency while preserving perceptual textures. Specifically, WEAVE routes style supervision to high-frequency subbands and applies a latent Sliced Wasserstein Distance alignment only at the final step.

We further introduce an identity-aware evaluation protocol that exposes cases where conventional metrics favor trivial identity mappings. Under this protocol, WEAVE consistently outperforms the identity baseline. Remarkably, with only 903K parameters, it trains in just 3 minutes on a single consumer GPU, making high-quality style transfer both effective and affordable.

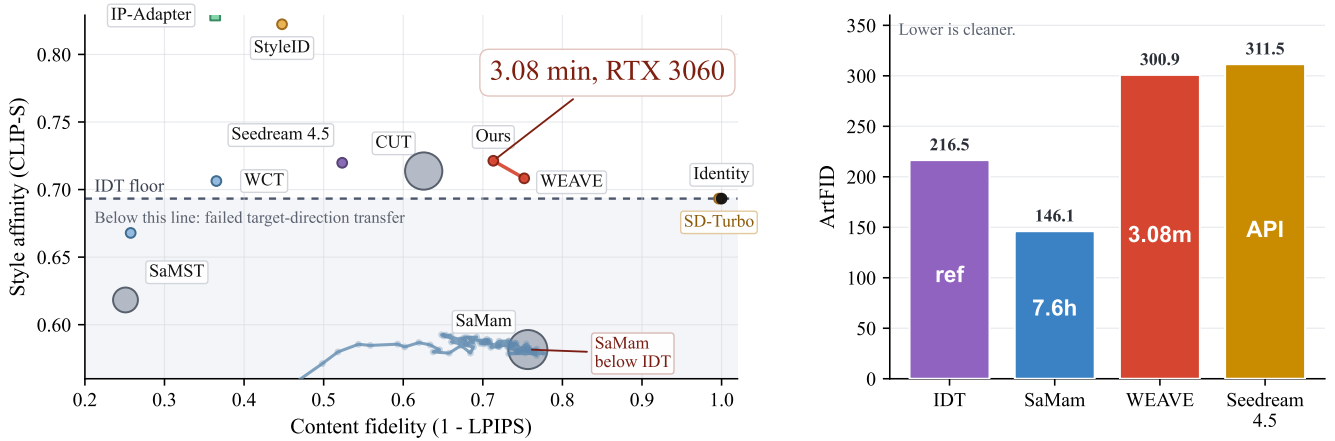


Figure 1: IDT-calibrated comparison on Distinct5-WikiArt. Left: CLIP-S versus $1 - \text{LPIPS}$, with bubble area encoding training time. The red points mark two WEAVE operating points. Square markers denote training-free diffusion editors (StyleAligned, IP-Adapter). The dashed line marks the IDT floor (0.6933); methods below it have not moved in the requested direction. Right: target-pooled ArtFID on the same benchmark, with numbers inside the bars indicating training or response time.

Introduction

Style transfer currently faces a polarized dilemma in model deployment and evaluation. On one hand, state-of-the-art diffusion models (e.g., Qwen-Image and FLUX.2) and their training-free variants demand more than 60 GB of GPU memory, rendering them impractical for consumer hardware. On the other hand, lightweight alternatives that avoid these heavy generative backbones often struggle to achieve gen-

uine style transfer. Specifically, the style affinity of outputs from many lightweight models often falls below that of the unchanged source image (i.e., the identity mapping). This critical issue has long been masked by conventional evaluation protocols, such as LPIPS and ArtFID, which heavily reward structural reconstruction but rarely include the identity mapping as an explicit baseline. As visualized in Figure 1 (Left), when the identity mapping is explicitly plot-

ted, modern lightweight methods like SaMam clearly fall below this natural baseline. To expose this evaluation blind spot, we introduce the Identical-Image Transfer (IDT) calibration, transforming standard metrics into direction-aware measures.

Why do modern lightweight architectures, despite their advanced sequence-modeling capabilities, consistently fall into this identity trap? We argue that the root cause is not a lack of model capacity, but an inherent geometric conflict within the standard latent space. In typical flow-matching frameworks, the model is tasked with predicting a velocity field that transports the source latent toward the target. However, natural images follow a $1/f$ spectral distribution, meaning the low-frequency structural energy (e.g., global layouts and object shapes) is orders of magnitude stronger than the high-frequency stylistic textures (e.g., brushstrokes and material patterns). When forced to minimize the reconstruction loss in a unified latent space, the optimizer naturally prioritizes fitting the massive low-frequency errors. Consequently, the subtle style-conditioned gradients are completely overshadowed, driving the model to safely converge toward a style-agnostic identity mapping. This physical reality explains why simply scaling up parameters fails: structure and style are fundamentally competing in the same coordinate basis.

To solve the conflict between structure and style, we propose WEAVE. First, we must choose where to learn this transfer. Training a small model directly on raw pixels is slow and easily distracted by meaningless pixel noise, while full diffusion models provide clean features but demand massive GPU memory. We choose a practical middle ground: we use the pre-trained VAE from diffusion models to compress images into a clean, low-dimensional space, but we completely discard the heavy diffusion network itself. Instead, we build a tiny network (only 903K parameters) to learn the style transfer within this clean space.

Within this space, we apply a wavelet transform to separate the features into structure (low frequency) and texture (high frequency), which prevents the strong structure from hiding the subtle style. However, simply matching the overall texture often causes obvious errors, such as blending sky patterns onto human skin. To prevent this, we use the low-frequency structure to group similar object regions and match textures only within the same region (via Sliced Wasserstein Distance). Finally, to prevent the original content from being gradually destroyed during the generation steps, we apply this texture matching only at the very last step. Practical deployment demands not only efficient inference but also affordable retraining when adapting to new artistic domains. This streamlined design allows WEAVE to clear the identity baseline and finish training in just 3 minutes on a single consumer GPU. Furthermore, because our alignment mechanism operates directly on the latent features, it transfers seamlessly as a training-free approach to large pre-trained models.

Contributions. (1) **Evaluation Protocol:** We introduce the Identical-Image Transfer (IDT) calibration, exposing the blind spot where lightweight models underperform the unchanged source. (2) **Representation & Methodology:** We

reveal the frequency-dominance failure mode in shared latent spaces and propose WEAVE, a wavelet-decoupled framework that achieves genuine style transfer with only 903K parameters. (3) **Efficiency & Generalization:** WEAVE trains in just 3 minutes on a single consumer GPU, and its latent operations serve as a model-agnostic, training-free plugin for large diffusion backbones.

Related Work

Evolution of Style Transfer

Early style transfer relied on optimizing pixel spaces (Gatys, Ecker, and Bethge 2015) or matching feature statistics in pre-trained discriminative networks, such as AdaIN (Huang and Belongie 2017) and WCT (Li et al. 2017). Subsequent works introduced GANs and contrastive learning for unpaired translation, notably CUT (Park et al. 2020). Diffusion models further shifted style transfer from explicit feature transformation to generative editing in latent spaces. Recently, the field has been dominated by heavy generative priors based on latent diffusion models (Rombach et al. 2022). State-of-the-art backbones like FLUX.2 and Seedream, along with training-free attention manipulation techniques like StyleID (Chung, Hyun, and Heo 2024) and StyleAligned (Hertz et al. 2024), achieve strong style transfer performance by leveraging massive pre-trained representations.

Lightweight Architectures and Evaluation Protocols

To reduce computational costs, modern lightweight architectures, such as Mamba-based SaMST and SaMam (Liu et al. 2024, 2025), have been explored. While these methods improve efficiency, standard evaluation protocols (Zhang et al. 2018; Wright and Ommer 2022) heavily reward source reconstruction and historically omit the identity mapping as an explicit baseline. These observations highlight the importance of explicitly considering identity mappings when evaluating style transfer.

Spectral Bias and Frequency-aware Optimization

Neural networks exhibit a well-documented spectral bias (Rahaman et al. 2019), preferentially fitting low-frequency signals. Combined with the $1/f$ spectral distribution of natural images, this observation motivates frequency-aware optimization strategies.

Wavelet and Latent-space Alignment

Wavelet transforms have been widely adopted in image restoration and generative modeling to separate multi-scale information, and have been adapted for style transfer (Yoo et al. 2019) and diffusion models (Phung, Dao, and Tran 2023). Concurrently, latent diffusion models (Rombach et al. 2022) introduced perceptual latent spaces that compress high-dimensional pixels into regularized representations. This regularized distribution has facilitated statistical alignment methods such as whitening-and-coloring transforms (Li et al. 2017) and sliced Wasserstein distance, making distribution matching more stable in latent spaces.

Method

This section presents WEAVE’s architecture and its motivation. We first calibrate evaluation with identical-image transfer (IDT), then describe the wavelet transport, endpoint alignment, and distribution-level SWD guide used during training.

Problem Formulation

Let $x_c \in \mathbb{R}^{H \times W \times 3}$ be a content image and $s \in \{1, \dots, K\}$ a target style domain. We work in the latent space of the Stable Diffusion v1.5 EMA VAE. We write $z_0 = \mathcal{E}(x_c) \in \mathbb{R}^{4 \times 32 \times 32}$. During training, $z_1 \sim p_s$ is a latent sampled from the target domain. Rectified flow uses the straight segment $z_t = (1 - t)z_0 + tz_1$, $u_t = z_1 - z_0$, $t \sim \mathcal{U}([0, 1])$. At inference we use an 8-step Heun solver (Heun gave the best cost-quality tradeoff in our controls; Euler loses style, RK4 adds cost without improving the final result). We then decode $\hat{x}_s = \mathcal{D}(\hat{z}_1)$.

The Identity Transfer (IDT) Calibration

Style transfer is susceptible to a no-op shortcut: when the source already contains stylistic features, standard metrics can reward the identity mapping. We formalize this through identical-image transfer (IDT), where the unchanged source defines a no-op floor that any valid method must clear.

Definition 1 (Identical-Image Transfer (IDT) Floor). *Because the source image is already stylized, $\text{CLIP}(x_{\text{src}}, s_{\text{tgt}})$ is a natural no-op baseline. A valid transfer must satisfy $\text{CLIP}(x_{\text{out}}, s_{\text{tgt}}) \geq \text{CLIP}(x_{\text{src}}, s_{\text{tgt}})$; below-floor scores indicate negative transfer—the method has not moved in the requested direction.*

The IDT floor turns standard metrics into direction-aware measures: once the identity row is included, low LPIPS alone is not enough—a method must also clear the floor, or it is merely reconstructing the source. On Distinct5-WikiArt, SaMam achieves strong content fidelity (LPIPS 0.2434) but its CLIP-S of 0.5816 falls below the IDT floor of 0.6933—a signature of source reconstruction rather than target-direction transfer. IDT thus exposes an evaluation blind spot: when the source already contains rich stylistic features, the identity mapping can score deceptively well.

Wavelet-Decoupled Transport

WEAVE first changes coordinates. A single-level Haar wavelet transform decomposes the $4 \times 32 \times 32$ latent into four orthogonal subbands: a low-frequency LL band and three high-frequency bands (LH, HL, HH), each of size $4 \times 16 \times 16$. The transform is orthogonal, so the Frobenius norm splits additively:

$$\|z\|_F^2 = \|\ell\|_F^2 + \|h_1\|_F^2 + \|h_2\|_F^2 + \|h_3\|_F^2. \quad (1)$$

We write $\pi(z) = \ell$ for the low-frequency projection. Due to the orthogonality of the Haar transform, modifying only the high-frequency subbands leaves the LL coordinate unchanged.

Empirical frequency probes. Across 5,000 random cross-style latent pairs, LL contains 69.5% of the squared

target-transport energy, while normalized style separation is strongest in HH (Figure 3). Together, these observations motivate weak LL supervision and greater emphasis on high-frequency appearance statistics.

Haar is selected for its exact orthogonality, compact spatial support, and negligible computational overhead; a single decomposition level suffices for the compact $4 \times 32 \times 32$ latent resolution.

The overall architecture is shown in Figure 2.

High-Frequency Routing

Let $\mathcal{W}(z_t) = (\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$ and $\mathcal{W}(u_t) = (u_\ell, u_1, u_2, u_3)$. The WEAVE model receives $(\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$ and predicts three velocity heads (v_ℓ, v_1, v_2) . The h_3 (HH) band has no learned velocity head; it stays in the state and is stylized only at the endpoint.

The spectral flow-matching term is

$$\mathcal{L}_{\text{impl}} = \mathbb{E}_t [\lambda_{LL} \|v_\ell - u_\ell\|_2^2 + \|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2], \quad (2)$$

with $\lambda_{LL} = 0.3$. WEAVE de-weights LL rather than eliminating it: some styles also shift coarse color tone, so LL remains a weak anchor.

The predicted one-step endpoint is $\hat{z}_1 = z_0 + \mathcal{W}^{-1}(v_\ell, v_1, v_2, 0)$. The complete active objective is

$$\mathcal{L} = \mathcal{L}_{\text{impl}} + 8\mathcal{L}_{\text{SWD}} + 0.1\mathcal{L}_{\text{edge}} + \mathcal{L}_{\text{low}}, \quad (3)$$

where $\mathcal{L}_{\text{edge}}$ is an ℓ_1 match between high-pass residuals and $\mathcal{L}_{\text{low}}$ is an MSE anchor between low-pass endpoints.

Backbone architecture. The velocity network consists of four residual blocks (width 64, $\sim 903\text{K}$ trainable parameters) with three separate prediction heads—one 1×1 convolution per supervised subband (ℓ, h_1, h_2). The h_3 (HH) band receives no velocity head and evolves passively along the rectified-flow trajectory. Style conditioning is injected through a memory module: each target style s owns a learned embedding $m_s \in \mathbb{R}^d$ ($d=64$), which is projected to keys and values for cross-attention. The attention query is assembled from the high-frequency subbands via a lightweight linear projection $Q[h_1; h_2; h_3]$, and the resulting features are fused into the residual-stream activations. Critically, the LL subband enters the backbone as a separate channel and interacts with style only through the shared residual blocks—never through the attention path—preserving the intended frequency separation.

Loss-level de-weighting is only half of the design. At inference we route style queries through the high-frequency channels: $q_{\text{routed}} = Q[h_{1,t}; h_{2,t}; h_{3,t}]$, $k = K(m_s)$, $v = V(m_s)$, where m_s is the style memory. During training, we use the high-frequency query with probability $p = 0.8$ and the full-latent query with probability 0.2, preserving a small low-frequency color channel while matching inference. We use RMSNorm (not GroupNorm) throughout: GroupNorm zero-centers each group, erasing the channel-wise mean that carries tonal information; in our runs it strips $\approx 72\%$ of style-relevant statistics, producing washed-out outputs.

The idealized limit $\lambda_{LL} = 0$ isolates high-frequency transport, but it is not the deployed setting. WEAVE retains weak LL supervision because coarse color and layout motion remain non-negligible in Figure 3.

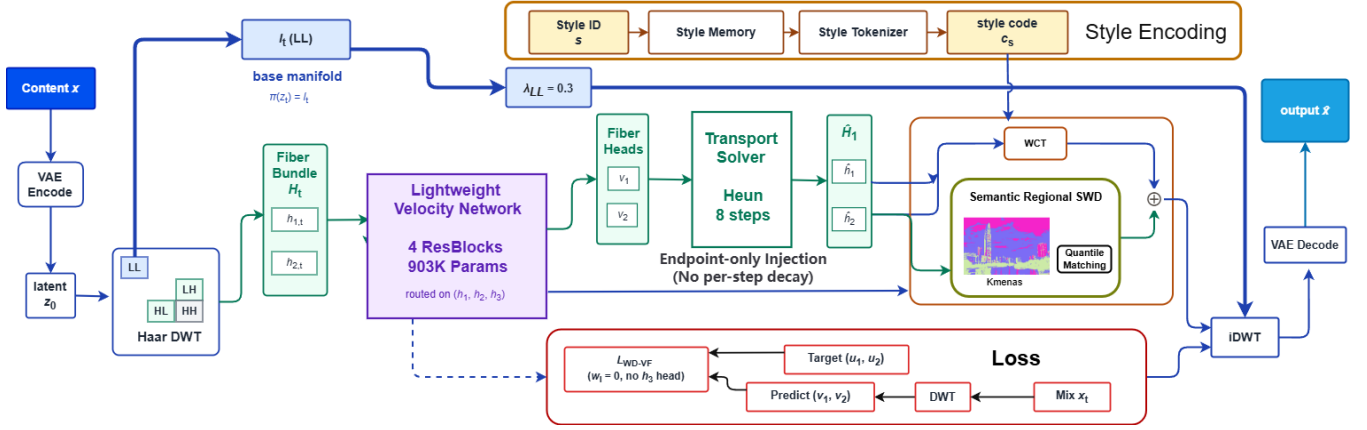


Figure 2: **WEAVE architecture.** A single-level Haar transform exposes LL/LH/HL/HH coordinates. The compact velocity network predicts LL/LH/HL motion, while style conditioning enters the shared backbone. Training combines band-weighted flow matching with an endpoint SWD guide; the current baseline uses its global, rather than semantic-region, form.

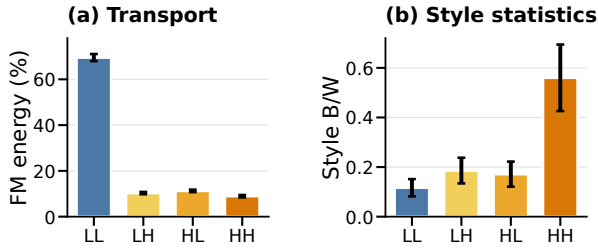


Figure 3: **Frequency probes.** (a) Cross-style transport energy in each Haar band. (b) Between-style versus within-style separation of latent statistics. Error bars summarize variation across style routes or statistics.

Endpoint Statistical Alignment

Rectified flow interpolates latents along straight segments, but linearly interpolating high-frequency wavelet coefficients between two images causes variance decay: the variance of two independent textures drops parabolically to a minimum at $t = 0.5$, yielding blurred, phase-misaligned results. Per-step style injection therefore compounds these artifacts, motivating endpoint-only statistical correction.

Exponential content decay from per-step injection. Per-step feature blending geometrically erodes the original signal: after n steps of blend strength α , only a fraction $(1 - \alpha)^n$ of the content survives (e.g., dropping to 3.9×10^{-3} for $n = 8, \alpha = 0.5$). WEAVE’s endpoint-only WCT sets $n = 1$, making the operation affine and thereby preserving structure identifiability independent of style strength.

Endpoint WCT. WEAVE applies whitening-and-coloring (WCT) only once, at the final endpoint. Let $\mathcal{W}(\hat{z}_1) = (\hat{\ell}, \hat{h}_1, \hat{h}_2, \hat{h}_3)$ and $\mathcal{W}(z_1^*) = (\ell^*, h_1^*, h_2^*, h_3^*)$. For $i \in \{1, 2, 3\}$, the whitening-coloring map is $T_i(a) = \Sigma_i^{*1/2} \hat{\Sigma}_i^{-1/2} (a - \hat{\mu}_i) + \mu_i^*$, where $(\hat{\mu}_i, \hat{\Sigma}_i)$ and (μ_i^*, Σ_i^*) are the mean and covariance of $\hat{h}_i$ and h_i^* . We set

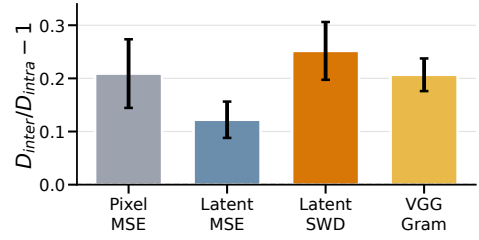


Figure 4: **Loss-level style separation.** Class-distance ratio on 150 Distinct5 images; zero means equal inter- and intra-style distance. Error bars are standard errors over ten style pairs.

$\mathcal{W}(\hat{z}_1^+) = (\hat{\ell}, T_1(\hat{h}_1), T_2(\hat{h}_2), T_3(\hat{h}_3))$ with endpoint scales $(0.0, 0.3, 0.3, 0.5)$. Each T_i operates only on its own sub-band, so $\hat{\ell}$ is never touched—global structure is preserved. The SD1.5 VAE’s KL regularization makes the latent approximately Gaussian (KL < 0.05 nats per subband), justifying second-order moment matching.

Sliced-Wasserstein Guide

Point-wise flow matching does not explicitly match the spatial feature distribution of the predicted endpoint. We therefore compare $\hat{z}_1$ and the sampled target latent z_1 with a global SWD. For 64 fresh random channel directions d_j , we project every spatial latent vector, sort the projected samples, and average their absolute quantile difference:

$$\mathcal{L}_{\text{SWD}} = \frac{1}{64} \sum_{j=1}^{64} \left\| \text{sort}(\hat{Z} d_j) - \text{sort}(Z_1 d_j) \right\|_1. \quad (4)$$

Here $\hat{Z}, Z_1 \in \mathbb{R}^{HW \times C}$ flatten the spatial dimensions. The directions are resampled on every call. In the current closed baseline this loss is global: semantic-region blending is disabled, and neither attention nor DWT-energy weighting is

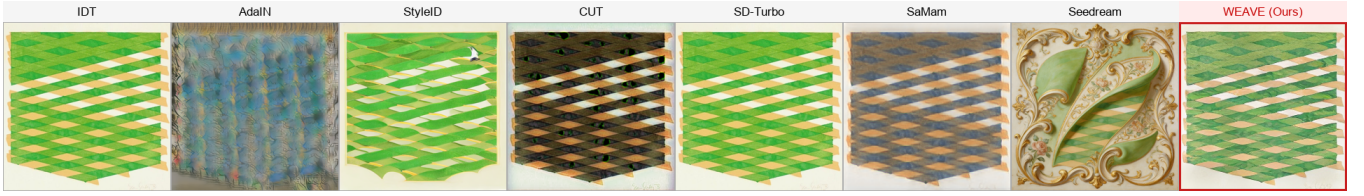


Figure 5: Extreme cross-domain stylization (Minimalism $\rightarrow$ Rococo). Heavy diffusion priors (Seedream, StyleID) hallucinate content, while classical methods and SaMam collapse to near-identity. WEAVE preserves source topology via low-frequency anchoring ($\mathbf{v}_{LL} \equiv 0$) and injects the target’s painterly textures exclusively through high-frequency subbands.

active. Thus SWD supplies a distribution-level style guide complementary to the paired spectral FM target, without claiming semantic correspondence.

The probe in Figure 4 explains why a distributional loss is useful in this latent space. Point-wise MSE loses style separation after EMA-VAE encoding, whereas latent SWD restores a class-distance margin comparable to pixel-space and VGG-statistic losses.

Mechanistic Interpretation

While analyzing global convergence in highly non-convex diffusion and flow matching models remains an open challenge, we can characterize this failure mode through a mechanistic lens based on the empirical spectral properties of natural images. If style transfer is learned directly in a shared latent space, the natural objective is

$$\mathcal{L}_{\text{euc}} = w_f \mathcal{L}_{\text{FM}} + w_s \mathcal{L}_{\text{sty}}, \quad (5)$$

where $\mathcal{L}_{\text{FM}} = \mathbb{E}_t \|\tilde{v}_\theta(z_t, t, s) - u_t\|_2^2$ and $\mathcal{L}_{\text{sty}}$ is any endpoint style-matching term. The difficulty is structural: the straight target $u_t = z_1 - z_0$ mixes low-frequency layout motion with high-frequency texture motion in the same coordinates.

Definition 2 (The Style-Suppression Regime). *We characterize the “no-op” failure mode through a regime Θ_0 where the velocity field largely ignores style conditioning: $\Theta_0 = \{\theta : \tilde{v}_\theta(z, t, s) \approx \tilde{v}_\theta(z, t)\}$ with $\tilde{v}_\theta = \mathbb{E}_s[\tilde{v}_\theta]$. Empirically, models operating near this regime exhibit strong content preservation but fall below the IDT style floor.*

Observation 1 (Frequency-Dominance in Gradient Signals). *Natural images and their VAE latents follow a $1/f$ spectral distribution. Consequently, the flow-matching loss $\mathcal{L}_{\text{FM}}$ enforces massive gradient penalties on low-frequency structural deviations ($L_f^{\text{low}} \gg L_f^{\text{high}}$), while style metrics (like CLIP) are relatively insensitive to high-frequency textures ($L_s^{\text{high}} \approx 0$). In a shared latent basis, optimization prioritizes minimizing the dominant low-frequency flow loss, heavily suppressing the style-conditioned high-frequency gradients. This bias encourages optimization toward the style-agnostic regime Θ_0 .*

WEAVE addresses this through wavelet decoupling: the DWT isolates low-frequency structure from high-frequency texture, reducing the coupling between structural and textural optimization. The ablation results (Section) are consistent with this mechanistic picture.

Few-Shot Generalization via a Decoupled Style Subspace

The high-frequency routing ($q_{\text{routed}} = Q[h_1; h_2; h_3]$) means the style query only sees the subbands (h_1, h_2, h_3) , which together with the style-memory embedding m_s form a **style subspace** that is approximately decoupled from the content (LL) subspace. The flow backbone thus learns a transport operator that is *style-agnostic*: s only enters through the high-frequency query path. When a new style is introduced, the transport dynamics stay valid; the only unknown is the target distribution of (h_1, h_2, h_3) , so few-shot adaptation reduces to estimating its mean and covariance from a few references—a low-dimensional problem, which is why 30 references suffice. Updating m_s does not interfere with content-preserving transport because the LL path is never touched by the style query.

Adaptation protocol. For a new style with K reference images, we extract their latents, decompose them via Haar, and compute the target subband statistics $\{(\mu_i^*, \Sigma_i^*)\}_{i=1}^3$ for endpoint WCT. The style-memory embedding m_s is initialized as the mean-pooled CLIP image feature of the references and is then optimized—together with the backbone parameters—on the standard WEAVE objective for 200 steps (approximately 20 seconds on an RTX 3060). We evaluate two modes: *training-free* (freeze the backbone, update only m_s and the endpoint statistics from 30 references) and *fine-tuning* (update the full model). The training-free mode already clears the IDT floor on all eight held-out styles; fine-tuning adds only a modest +0.008 CLIP-S at substantially higher compute cost, confirming that the decoupled geometry makes the content-to-style mapping largely reusable across unseen domains.

Experiments

Setup and Protocol

Setup. We use Distinct5-WikiArt-512 with five styles (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e) selected for having the lowest IDT CLIP-S, making accidental no-op success harder. Each domain has 3,600 training images and 150 test images; evaluation uses 750 source-target pairs, including 150 identity pairs for IDT calibration. For generalization, WEAVE is trained from scratch on all 20 WikiArt families and evaluated on all $20 \times 20 \times 30 = 12,000$ source-target pairs; baselines are evaluated on the same Distinct5 subset (750 pairs) to keep compute bounded. CLIP-S (ViT-B/32) and LPIPS (AlexNet) are the

parameters, and the entire system runs on a single consumer GPU.

Auxiliary metrics agree with the main reading. Figure 1(right) shows target-pooled ArtFID of 300.9 for WEAVE versus 311.5 for Seedream 4.5, placing both in the same realism range; on all-pairs evaluation WEAVE reaches 0.7213 CLIP-S / 0.2868 LPIPS, confirming the conclusion under the ablation protocol.

Mechanistic Ablations

Our subtractive audit shows that modules standard in prior work—optimal-transport routing, multi-step guidance, DINO correspondences—are not only unnecessary but often harmful here: Observation 1 indicates that any component coupling structure and texture in one basis is pulled toward the style-suppression regime. We strip them, leaving a minimal core the optimizer can use for target-direction transport.

The 512×512 audit spans 43 configurations across seven theoretical axes (AdaIN 4× retained in the CSV but omitted from the main text for space).

Most configurations cluster tightly around the full model, confirming the robustness of WEAVE’s core design. The most prominent off-frontier point is the extrapolation 1 variant, whose content preservation collapses when high-frequency routing is removed, validating the role of endpoint subband alignment. A separate 10× learning-rate run diverged to NaN and is excluded from the plot. The AdaIN 4× variant was also omitted for readability because it caused severe structure distortion.

Axis-by-axis summary. The seven axes are LL weight $\lambda_{LL} \in \{0, 0.1, 0.3, 0.5, 1.0\}$ (0 removes low-frequency supervision), routing probability $p \in \{0.5, 0.8, 1.0\}$, endpoint scale vector, WCT on/off, RMSNorm vs. GroupNorm, residual blocks (2 vs. 4), and width (32 vs. 64). Across 43 configurations the full model is a clear Pareto optimum: no single-axis change improves both CLIP-S and LPIPS, and removing high-frequency routing ($p=0.5$) causes the largest CLIP-S drop (−0.032), while more blocks yield diminishing returns.

Why the cluster is tight. Most tested configurations cluster tightly, indicating moderate sensitivity to the scanned hyperparameters. We avoid interpreting this finite sweep as evidence of a globally flat or separable loss landscape.

Generalization and Robustness

Latent space vs. RGB. A matched 256×256 control shows the gain comes from wavelet transport in VAE latent space, not an extra loss: direct RGB flow reaches only 0.5842 CLIP-S / 0.4121 LPIPS, far worse than WEAVE, and the gap persists even at matched training time and parameters.

Few-shot style adaptation. Because WEAVE decouples content (LL) from style injection (LH/HL/HH), the high-frequency transport is not tied to a frozen vocabulary. Adapting on 8 held-out styles gives 0.7039 CLIP-S / 0.2555 LPIPS. Freezing the backbone and updating only the style-memory (30 images/style) already clears the IDT floor on all 8; fine-tuning the backbone adds only +0.008 CLIP-S at higher cost.

Random5-WikiArt generalization. To rule out cherry-picking, we train WEAVE from scratch on all 20 WikiArt families (≥ 530 images each) and evaluate all 12,000 pairs, obtaining 0.7434 CLIP-S / 0.2904 LPIPS and clearing the Random5 identity floor of 0.7312 with LPIPS < 0.30. On the 750-pair 5-style subset, the baseline WEAVE reaches MUSIQ 44.24; applying semantic region SWD plus EOTA raises MUSIQ to 53.73 at 0.7182 CLIP-S / 0.3735 LPIPS, confirming the semantic-SWD mechanism generalizes beyond Distinct5. StyleID reaches the highest raw CLIP-S (0.8180) but at 0.5827 LPIPS; SD-Turbo and CUT lie between the high-style/high-distortion and low-style/low-distortion regimes, while SaMST stays below the identity floor (0.6669). The per-family breakdown shows consistent positive-IDT transfer everywhere, weakest on Pop Art and strongest on Ukiyo-e and Expressionism.

Wavelet basis. Daubechies-2 slightly raises CLIP-S but worsens LPIPS from 0.2868 to 0.3288. We retain Haar for its exact orthogonality and local support, which keeps both the theory and implementation simple while effectively avoiding blocking artifacts.

Resolution robustness. A 512-trained model evaluated on 256×256 inputs via inference-time downsampling (no retraining) drops only −0.018 CLIP-S and +0.031 LPIPS, showing the subband transport transfers to lower resolutions; upsampling a 256-trained model to 512 is far worse (−0.047 / +0.069), confirming high-frequency bands benefit from native-resolution training.

Discussion

Limitations. WEAVE is domain-conditional (a style family, not a reference painting). The Distinct5 benchmark is intentionally narrow (five hard, low-IDT styles as a no-op stress test), partially addressed by the Random5-WikiArt extension; the LL-protecting design makes palette-heavy targets like Minimalism harder than textured ones. The theory is local and mechanism-oriented; broader cross-domain evaluation would strengthen the empirical case.

Conclusion

Style transfer needs both the right success criterion and the right coordinate system. IDT calibration exposes when a method merely reconstructs the source, and WEAVE avoids this regime by separating low-frequency structure from high-frequency style motion and routing style through the same high-frequency path used at inference. The 903K-parameter backbone trains in 3.08 minutes on one RTX 3060. The D5 four-metric result is withheld until the same-checkpoint evaluation is complete; the current probes support band-aware modeling without claiming a complete content–style factorization. The central finding: real target-direction transfer does not require a large model once the transport geometry separates structure from style, and by running on a single consumer GPU WEAVE lowers the barrier for high-quality stylization.

Ethics Statement

Style transfer tools can be misused to produce deceptive images or to misattribute stylistic authorship. Our method operates at the domain level (e.g., “Impressionism”) rather than the per-artist level. This reduces the risk of imitating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt and contains historically significant artworks. It does not contain personal data. We encourage downstream users to disclose when an image has been style-transferred and to respect source-content licensing terms.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). The experiments section gives the training configuration, hyperparameters, and evaluation protocol. Training takes 3.08 minutes on a single RTX 3060 (12 GB). The 750-pair evaluation run needs 83.7 seconds for generation plus VAE decode and 97.3 seconds including CLIP/LPIPS metrics. The 903K trainable parameter backbone checkpoint is 3.5 MB. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.